

Hongkang Chu

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Multimodal learning; AI agents; synthetic image and video detection; MRI and MRSI reconstruction

Education

University of the Chinese Academy of Sciences 2025–present

Master’s student, Computer Science. Expected graduation: 2028.

Shanghai Ocean University 2021–2025

Bachelor’s degree, Computer Science and Technology. Undergraduate research in protein language models.

Research Experience

Chinese Academy of Sciences / UCAS 2025–present

Graduate Research

Research in MRI, MRSI, and deuterium metabolic imaging at the Innovation Academy for Precision Measurement Science and Technology.

- Co-developed prior-informed deep learning reconstruction for weighted-average CSI, combining anatomical and spectral information; first-author ISMRM 2026 Power Pitch Oral.
- Co-authored work on physics-embedded CycleGAN reconstruction for short-readout deuterium metabolic imaging, published in Magnetic Resonance Letters.
- Co-authored a phase model-driven deep learning study for NMR phase correction, published in The Journal of Physical Chemistry Letters.

Zhejiang University Apr–Sep 2026

Research Collaboration

Collaborative research in detecting AI-generated images and videos.

- Second author of TRACE, studying generative-model velocity responses and temporal consistency for synthetic video detection; submitted to ICLR 2027.
- Second author of ClueAegis, using perceptual clues, forensic skill selection, and evidence-guided reasoning for synthetic image detection; accepted to Findings of EMNLP 2026.

University of Illinois Urbana-Champaign Feb 2026–present

Remote Research Intern

Research in multimodal models and medical AI.

Shanghai Ocean University 2021–2025

Undergraduate Research

Protein language models for druggable protein classification.

- Implemented ESM-2 embedding classifiers and an adapted GPT-2 sequence classifier, with additional evaluation on Pharos; first-author IJMS publication in 2024.

Technical Skills

Python and PyTorch; Multimodal learning and transformer fine-tuning; Tool-using agents and model evaluation; Diffusion and flow-matching representations; MRI / MRSI reconstruction.

Selected Publications and Conference Contributions

TRACE: Trajectory Representation and Consistency Estimation for AI-Generated Video Detection

Huangsen Cao, **Hongkang Chu**, Siyao Yu, Xin Ding, Jianfeng Dong, Yongwei Wang.
ICLR 2027 Conference Submission. Under Review. Second Author.

ClueAegis: Heuristic-to-Reasoning Cognitive-skill Learning for Unified Evidence-based Synthetic Image Detection

Huangsen Cao, **Hongkang Chu**, Yuxi Li, Ying Zhang, Chen Li, Jing Lyu, Yongwei Wang, Yu Zhao, Fei Wu.
Findings of EMNLP 2026. Accepted. Second Author.

Physics-embedded CycleGAN for robust metabolite mapping in short-readout deuterium metabolic imaging

Shuang Huang, Gang Chen, Xinjie Liu, **Hongkang Chu**, Hong Shi, Zhi Zhang, Xin Zhou, Maili Liu, Chaoyang Liu, Qingjia Bao.
Magnetic Resonance Letters, 6(4): 200272, 2026. Published Online. Co-author.

Phase Model-Driven Deep Learning for Robust Phase Correction in High-Throughput NMR-Based Metabolomics

Chuanwen Zhao, Gang Chen, Caixiang Liu, Zhao Li, Jing Zhao, Peng Sun, **Hongkang Chu**, Xin Zhou, Maili Liu, Peijun Song, Qingjia Bao, Chaoyang Liu.
The Journal of Physical Chemistry Letters, 17(4): 1245–1253, 2026. Published. Co-author.

Deep Learning Enhanced Weighted Average CSI for High-Resolution Deuterium Metabolic Imaging

Hongkang Chu, Gang Chen, Xinjie Liu, Hong Shi, Chaoyang Liu, Maili Liu, Xin Zhou, Elton Montrazi, Lucio Frydman, Qingjia Bao.
ISMRM 2026, Cape Town, South Africa; 11 May 2026. Power Pitch Oral. First Author.

Comprehensive Research on Druggable Proteins: From PSSM to Pre-Trained Language Models

Hongkang Chu, Taigang Liu.
International Journal of Molecular Sciences, 25(8): 4507, 2024. Published. First Author.